Economic Complexity – Homework 3 – Petra Hilbert and Natália Szabó

Former coworkers and low-paid workers' moves into better-paid jobs in Hungary

1. Problem and gap

Many low-paid workers in Hungary stay in low-paid jobs for years. One way out is to move to a better-paid industry where their skills are still useful. Previous research shows that when workers change jobs, they mostly move between industries that need similar skills (Neffke, Otto & Weyh, 2017). Evidence from Sweden also shows that low-paid workers are more likely to move into better-paid jobs in regions where better-paid jobs in industries that require similar skills are available (Elekes, Baranowska-Rataj & Eriksson, 2023).

People move into jobs where they can utilize their existing skills, but personal connections, such as former coworkers, may help them find and access those jobs. Studies of coworker networks show that people who lose their job find a new one faster when more of their former coworkers are employed (Glitz, 2017; Cingano & Rosolia, 2012), and the effect is stronger when these contacts work at firms that are close to the worker's old job in location and technology (Cingano & Rosolia, 2012). Firms also like to hire former coworkers of their own employees, because their staff can tell them who is a good worker (Hensvik & Nordström Skans, 2016).

Gap statement: Previous work has shown that workers mostly move between skill-related industries and that former coworkers help people find new jobs, but we examine whether these contacts help low-paid workers in Hungary move into better-paid, skill-related industries, beyond the opportunities available locally. This matters because if contacts, not only skills, decide who moves up, then training alone will not be enough, and policy also needs to connect low-paid workers with people in better-paid jobs.

2. Research question and hypothesis

2.1 Research question(s)

Are low-paid workers more likely to move into a better-paid, skill-related industry if some of their former coworkers already work there?

Potential additional questions:

- Does knowing someone matter more than simply having many such jobs nearby?
- Which skills and which policy tools (training, education, investment) can help low-paid workers make this move, especially those without contacts?

2.2 Relevance of networks

Skill relatedness tells us where a worker could go and the coworker network may decide whether and how she actually gets there. As we saw in the existing research, one of the networks alone is not sufficient to answer our research question, but the two channels can work together to predict worker movement.

2.3 Hypotheses:

H1: Low-paid workers with at least one former coworker in a better-paid, skill-related industry are more likely to move into such an industry within two years.

H2: This coworker effect stays strong even when we control for how many better-paid, skill-related jobs there are in the region.

H3: Contacts matter less for workers with a vocational or higher degree, because a formal qualification can partly replace a personal recommendation.

3. Data and method

3.1 Data

We use the Admin4 database of the KRTK Databank: linked administrative data on a 50% random sample of the Hungarian population, 2003-2021 (Boza, Pető & Tir, 2026). It contains monthly jobs with anonymised employer IDs, industries, occupations, wages and education. We focus on 2011-2020, when industry codes are available for all firms. We call a worker low-paid if her wage is in the bottom 20% of the wage distribution in a given year.

3.2 Nodes and links

Industry network: nodes are industries (4-digit NACE). two industries are linked (skill-related) if more workers move between them than we would expect by chance.

Coworker network: nodes are workers, two workers are linked if they worked at the same firm in the same month. (We only use firms with fewer than 50 employees, where people are likely to know each other, and keep a tie for five years after the two people stop working together.)

3.3 Network measures
For each low-paid worker we calculate:

Bridge ties: the number of former coworkers who now work in a better-paid, skill-related industry.
Local relatedness density: the share of jobs in the region that are in better-paid, skill-related industries
Network segregation: the share of the worker's former coworkers who are also low-paid.

3.4 Method

The outcome is whether the worker moves into a better-paid, skill-related industry within two years. We regress it on bridge ties and local density, with controls for age, gender, education, occupation, region and year. We compare workers who left the same firm in the same year as they had the same job and the same coworkers there, but their earlier coworkers ended up in different places to control for the possibility that people with good contacts may simply be better workers.

3.5 Data accessibility

Admin4 is available to researchers on application, through a remote server of the KRTK Databank. Approval takes time.

4. Expected results and impact

If our hypotheses hold, low-paid workers with a former coworker in a better-paid, skill-related industry will be clearly more likely to move up than similar workers from the same firm without such a contact, and this effect will be at least as large as the effect of having many such jobs nearby. We also expect the effect to be largest for workers without a vocational or higher degree, and workers whose former coworkers are mostly low-paid to move up the least. This would matter for several groups:

• Low-paid workers: it would show that being “stuck” is often not only a skill problem but also a contact problem.

• National Employment Service and county employment offices: training could be combined with ways to build contacts - mentoring by people already in better-paid jobs, job shadowing, short work trials at firms in skill-related industries, and help with referrals.

• Education and adult training: the “bridge skills” we find (the occupations and qualifications of successful movers) show which short courses and vocational certificates are worth funding. A recognised certificate could partly replace a personal recommendation.

• Investment policy: when the government or a county attracts a new plant (for example in car or battery manufacturing), it can check whether local low-paid workers are in skill-related industries and support local and referral-based hiring, so the new jobs also reach them.

• Firms with labour shortages: they can reach reliable workers through their own staff's networks (for example with referral bonuses), which is cheaper than open recruiting. Overall, the goal is to help fewer people stay stuck in low pay and to help regions make better use of the skills they already have.

References

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